



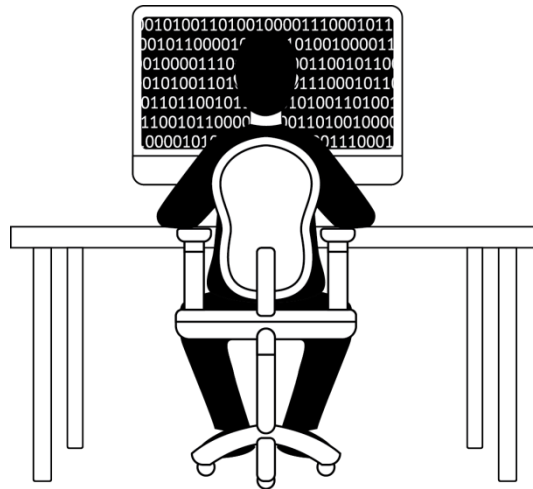
Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points

**Let's Start!****Integer Data Types****Learning Outcome:**

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- What do you think are some of the most used functions of the integer data units? List and share some of them.
- According to you, what is the key to understand and comprehend the coding field? Why?

Can Do Statements:

- I can show how to use syntax for short, int, and long data types.
- I can give detailed information about the short data type in Java.
- I can present information about the int and long data types in Java.



Let's Explore!

Getting Ready for the Exam



Instructions: Read the text below.

- Diego:** I'm a bit nervous about tomorrow's exam. I heard that not many students tend to pass it.
- Tara:** Well, I was one of the few that passed it last year. Indeed, you'd better study thoroughly as the teacher might want to get full answers on all term's topics. One of the topics that not many students check is the syntax in integer data types; so, what can you tell me about that?
- Diego:** The integer, which contains the short, int, and long data types, presents numeric values that are whole values, large numbers. Now, the variable declaration of these data types follows a specific syntax in which we assign a value to the variable. In other words, you might want to code something like `variable type variable name = 15;`
- Tara:** Correct. You also ought to make an in-depth review of short data types.
- Diego:** Let me see. The short data type is the smallest one. Plus, it takes 2 bytes of memory which is the same as 16 bits. It also shares the characteristic of accepting negative as well as positive values, just like the int and long data types do. Moreover, its default value is zero and you ought to only introduce values within the range of -32,768 to 32,767.
- Tara:** Great! You might want to also check the int and long data types...
- Diego:** Okay, I'd better recall that the int type consumes 32 bits of memory and that it takes values from -2,147,483,648 to 2,147,483,647. Regarding the long one, it takes 64 bits of memory and you ought to introduce values from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807.
- Tara:** Now, you'd better memorize all of that before you present the exam as not many students really prepare for it. Don't forget, you ought to pass that exam!

**Let's Practice!**

Remember this for the term exam.

Instructions: Circle the correct word to complete the sentence.

1. You _____ the use of syntax to presents numeric values thoroughly.
 - a. had better study
 - b. had studied
 - c. had better to study
2. You'd _____ the data types that it contains, which are the short, int, and long ones.
 - a. better memorizes
 - b. better are memorizing
 - c. better memorize
3. The teacher might _____ full answers on syntax, which are whole values, large numbers.
 - a. want to get
 - b. wants to get
 - c. is getting
4. You had _____ that we always assign a value to the variable.
 - a. better remember
 - b. better remembers
 - c. Remembered
5. You _____ something like: variable type variable name = 15;
 - a. ought to code
 - b. need to read
 - c. codes

**Let's Practice!**

Reviewing Different Data Types.

Instructions: Proofread the sentences. Correct the mistakes.

1. They had better made an in-depth review of short data types.
2. You have ought know that the short data type takes 2 bytes of memory.
3. You might want understand that it shares the characteristic of accepting negative and positive values.
4. You have better have in mind that its default value is zero.
5. You ought to only introducing values within the range of -32,768 to 32,767.





Extension Activity

Expanding the Acquired Knowledge.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for extra information about short, int and long data types.
- Write down some notes and create a brochure where you explain main features and its relevance in programming.
- Share with your partners and answer any question they might have.

Self-Assessment

Yes

No

- I look for extra information about short, int and long data types.
- I create a brochure to summarize the features and relevant facts.
- I talk about short, int and long data types.

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Target Words

- **thoroughly** (adverb): in a complete, conscious and deep way
e.g. Indeed, you'd better study thoroughly as the teacher might want to get full answers on all term's topics.
- **whole value** (noun): a whole value is a complete one that has no decimals
e.g. The integer, which contains the short, int, and long data types, presents numeric values that are whole values, large numbers.
- **assign** (verb): to provide something with a specific value
e.g. Now, the variable declaration of these data types follows a specific syntax in which we assign a value to the variable.
- **in-depth** (adjective): covering as much detail and information as possible
e.g. You also ought to make an in-depth review of short data types.
- **default** (noun): standard
e.g. Moreover, its default value is zero and you ought to only introduce values within the range of -32,768 to 32,767.
- **recall** (verb): to remember something
e.g. Okay, I'd better recall that the int type consumes 32 bits of memory and that it takes values from -2,147,483,648 to 2,147,483,647.



Learning Points

Had Better for What Is Advisable

You use **had better** to share ideas with others regarding what you believe is the best action to take in a specific situation.

When you use **had better**, you are trying to give a recommendation about a present situation in which you are telling another person what to do. The use of **had better** suggests that if the other person does not follow what you propose, a bad consequence will happen.

To make sentences using **had better**, you have to follow the structure:

Subject + had better + verb (base form) + complement.

For example,

You **had better** check that the code is correctly written.

The university **had better** teach its students how to write codes.

Remember that you can also use the abbreviation of the expression **had better** which is **'d better**.

Examples

- *Indeed, you'd better study thoroughly as the teacher might want to get full answers on all term's topics.*
- *Now, you'd better memorize all of that before you present the exam as not many students really prepare for it.*

Quantifiers, Not Many

You use **quantifiers** to enunciate the quantity or amount of something.

The negative form of the quantifier **many** needs a countable noun right after the expression. It implies the idea that something is not frequently seen or done.

Examples:

Not many people know about coding and Java.

Not many codes are easily understandable.

Not many software technicians specialize on Java.

Examples

- *I heard that not many students tend to pass it.*
- *Now, you'd better memorize all of that before you present the exam as not many students really prepare for it.*

